

Pack a Multi-Item Order SOP

One episode packs one order into one carton. Four items go into the carton heaviest first, three crumpled kraft paper balls fill the voids around and over them, the pack is shake-tested, the top flaps are folded and taped down the center seam, a shipping label goes on the patch, and the carton is slid to the lane spot. The episode ends with a sealed, labelled carton on the lane spot and a bare staging board.

The order never changes: nest, fill, shake, tape, label, stage. Each stage starts only after the one before it is complete. Nothing goes into the carton after the shake test, and the carton is never opened again once it is taped.

The items are laid out in weight order before the episode starts, closest to the packing spot heaviest. So weight is never judged during a take — the staged order is the loading order, and the heaviest item ends up on the floor of the carton with the lightest on top.

Every void is filled the same way: the **far gap first, then the near gap**, then the top. Filling the near gap first forces the arm to reach out over it to fill the far one, which drags the near ball out of place.

The left arm holds and the right arm works. The left gripper pins the carton or braces a sheet of paper and keeps it still. The right gripper does every lift, every placement, every crumple, every fold, and every press. The left gripper never carries an item, a paper ball, the tape strip, or the label.

Everything the right arm picks up is staged on the **right**. The carton stands on the packing spot in the middle and is not moved off it until the last step, when both grippers slide it **left** to the lane spot.

The table is set up in one of three ways. Only the four items move; the carton, the staging board with its paper stack and peel board, the crumple spot, the lane spot, and the scrap cup are in the same place in all three.

- **Config M:** the items stand in a row at the back-center, behind the carton, the row running away from the carton with item 1 nearest the carton's far wall.
- **Config R1:** the items stand in a row on the staging board on the right, item 1 nearest the packing spot.
- **Config R2:** the items stand in a row at the back-right, behind the staging board, item 1 nearest the packing spot.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the items start at the back-center (**Config M**), on the staging board (**Config R1**), or at the back-right (**Config R2**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the **right gripper** takes every item in all three configs. The left side of the table is never a start zone, because the left gripper is pinning the carton and never carries an item; the back-center counts as the right gripper's side. No arm reaches across the table.
- **Fixed roles:** everything else is the same in all three configs — the left gripper pins and braces, the right gripper lifts, places, crumples, folds, peels, and presses, the paper stack and the peel board are always on the staging board on the right, and the carton stays on the packing spot until the slide.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the packing spot in the middle, the staging board on the right, the items in the start zone for this episode's config, the crumple spot in the near middle, the lane spot on the left, and the scrap cup on the near right.
3. The open top of the carton is fully in frame from above, so the floor, both gaps, all four items, and all three paper balls can be seen going in.
4. The **top face** of the carton is in frame, so the center seam, the tape strip, and the label patch are visible once the flaps are folded.
5. Both arms are at home with their grippers open.
6. The tabletop is clear of everything except the zones listed below.
7. The right gripper can reach the **far corners of the carton floor** without the arm leaning out over the carton. If it cannot, move the packing spot toward the front edge until it can.
8. The right gripper can reach the **furthest item in the row** — on the staging board (Config R1), at the back-right (Config R2), or at the back-center (Config M) — without the arm going to the end of its reach. If it cannot, move the row closer to the packing spot.
9. The left gripper can pin the **left wall of the carton** with the arm working on its own side of the tabletop. If it has to reach past the middle of the carton to do it, move the packing spot to the left until it does not.

10. Both grippers reach the **crumple spot** in the near middle without either arm crossing to the other side of the tabletop.
11. The two arms clear each other over the carton — test this with the left gripper on the left wall and the right gripper over the near gap.
12. The path from the packing spot to the lane spot is clear and flat, with nothing for the carton to catch on.

Materials checklist

1. One **carton** stands square on the packing spot, open end up, bottom already taped, empty. Its four top flaps are folded out flat so the opening is clear.
2. The carton opening is a rectangle with its long side running **left to right**.
3. Four **items**, sealed and undamaged, stand in a row in the start zone for this episode's config, numbered **1 to 4** with item 1 closest to the packing spot. They are laid out in weight order before the episode: **item 1 is the heaviest and item 4 is the lightest**.
 - **Config M**: back-center, behind the carton, the row running away from the carton
 - **Config R1**: on the item spots on the staging board on the right
 - **Config R2**: back-right, behind the staging board
4. Each item lies flat on its largest face in the start zone, the same way it will lie in the carton.
5. Item 1 covers the floor of the carton on its own, and no item is larger than the one before it, so the four make a stack with nothing overhanging.
6. The stack of four items leaves a clear **far gap** and a clear **near gap** between the load and the carton walls, both wide enough to take a paper ball.
7. The four items stacked, plus the top fill pressed down, leave the top of the load **below the carton rim**.
8. Four **kraft paper sheets** lie in a stack on the paper spot, one on top of the next, each free of tears and each large enough to make a ball that fills a gap. Three are used and one is the spare.
9. One **tape strip** and one spare lie on the peel board, sticky side down, **grab tab pointing right**. Each peels off the board cleanly in a single piece.
10. One **shipping label** and one reserve lie on the peel board, sticky side down, **grab tab pointing right**.
11. The **label patch** is printed on the long top flap to the **right of the center seam**, and it is clean, dry, and bare.
12. The **crumple spot** in the near middle of the tabletop is bare and flat, and one sheet lies flat on it without hanging over an edge.
13. The **lane spot** on the left is outlined, bare, and big enough for the whole base of the carton.
14. The **scrap cup** stands on the near right of the tabletop, empty and clear of the arm paths.

Workspace layout

- **Packing spot:** in the middle of the tabletop, centered between the two arms. The carton stands here for Steps 1 to 6 and does not move off it until Step 7.
- **Gaps:** the carton floor holds the load in the middle with a **far gap** along the far wall and a **near gap** along the near wall. The far gap is always filled first.
- **Start zone:** where the four items stand in a row at the start — back-center behind the carton (**Config M**), on the staging board (**Config R1**), or back-right behind the staging board (**Config R2**). It is bare when the episode ends.
- **Staging board:** on the right, within reach of the right arm. It holds, in the order they are used and closest to the packing spot first: the four items in a row (**Config R1** only), the paper stack, and the peel board with the tape strips and the labels. In **Config M** and **R2** the item spots are bare.
- **Crumple spot:** a bare flat patch in the near middle of the tabletop, between the packing spot and the front edge. Paper balls are made here and nowhere else.
- **Lane spot:** on the left, outlined on the tabletop. The finished carton is slid onto it at the end of the episode. Nothing else is ever staged there.
- **Scrap cup:** on the near right of the tabletop. It is the only place backing strips and scrapped labels are dropped.

Vocabulary

- **Arm assignment:** the left arm holds. It pins the carton, or braces a sheet at the crumple spot, and keeps it still. The right arm works. It does every lift, placement, crumple, fold, peel, and press, and takes every item from the start zone in all three configs. Neither arm takes over the other's job, and the left gripper never carries an item, a paper ball, the tape strip, or the label.
- **Reach rule:** the left arm touches the **left wall of the carton**, the sheet at the crumple spot, and the left wall during the slide. The right arm reaches the staging board, the whole carton, the crumple spot, the scrap cup, and the start zone (the back-center in **Config M**, the back-right in **Config R2**). Neither arm crosses to the other side of the tabletop.
- **Carton:** the shipping box that gets packed. Its bottom is already taped when the episode starts.
- **Item:** one sealed thing from the order. It is handled by its two long sides and always laid flat on its largest face.
- **Start zone:** where the four items stand in a row at the start of the episode — back-center behind the carton (**Config M**), on the staging board (**Config R1**), or back-right behind the staging board (**Config R2**). One per episode, chosen before recording and never changed mid-episode.

- **Weight order:** items 1 to 4 as they are staged, heaviest to lightest. They are loaded in that order, so the heaviest sits on the carton floor and the lightest sits on top. Weight is never judged during the episode — the staged order is the loading order.
- **Load:** everything already in the carton — the items that are in, plus any fill that is in.
- **Base item:** item 1, the heaviest. It goes in first, flat on the carton floor.
- **Gap:** one of the two voids between the load and a carton wall — the **far gap** along the far wall and the **near gap** along the near wall.
- **Top fill:** the third paper ball, pressed flat over the top of the load.
- **Paper ball:** one kraft sheet bunched at the crumple spot into a loose ball. It is loose enough to squash into a gap when pressed, and big enough that it does not drop straight to the bottom of the gap and leave the top of the gap empty.
- **Brace:** the closed left gripper presses down on a sheet at the crumple spot and holds it still while the right gripper bunches it. A brace stays in one place — it never slides along.
- **Pin:** the closed left gripper presses firmly on the carton's left wall to hold it still. A pin never pinches and never slides, and it is held for the whole step.
- **Lower in:** the right gripper carries the item or ball down into the carton until it touches what is under it, and only then opens. Nothing is ever let go above the carton.
- **Flat and level:** the item lies on its largest face with all of it touching what is under it. No corner is lifted, no edge rides up on a neighbour, and it does not rock when the gripper leaves.
- **Seated:** the base item lies against the carton floor along its whole face, with no corner riding up a wall and no part of it bridging across the floor.
- **Below the rim:** the whole top of the load sits under the top edge of the carton, so the flaps fold shut over it without being pushed down.
- **Rock:** the left gripper, closed, stands against the bottom of the carton's **left wall** so the carton cannot slide. The right gripper, closed, presses on the **right wall** near its top and pushes left until the carton tips up onto its left bottom edge, about a gripper's width of daylight under the right side. Hold for 1 second, then ease it back level. The left bottom edge never leaves the desk and the carton is never lifted.
- **Shake test:** two rocks in a row, followed by a look down into the open carton to see whether anything in the load has moved.
- **End flaps:** the two short top flaps, one at the left end and one at the right end. They fold in first and end up trapped under the long flaps.
- **Long flaps:** the two long top flaps, one at the far side and one at the near side. They fold in last and meet in the middle.
- **Center seam:** the line where the two long flaps meet, running left to right down the middle of the carton's top.

- **Fold:** the right gripper pinches a flap in the middle of its free edge, swings it in until it lies flat and level, holds it there for 2 seconds, then releases.
- **Strip:** the length of tape, already cut to size. It is only ever handled by its **grab tab**.
- **Grab tab:** the folded-back end of a strip or a label, which has no sticky side showing. It is the only part a gripper ever touches.
- **Hangs straight:** the strip or label dangles free and flat from the lifted grab tab, without curling, twisting, or sticking to itself. One that is not hanging straight is dropped in the scrap cup — do not try to straighten it.
- **Lay:** the right gripper lowers the free end of the strip onto the start of the seam, then travels along the seam so the strip settles onto it behind the gripper, and releases the tab at the far end.
- **Overhang:** the part of the strip that hangs past the end of the seam and gets pressed down onto the end panel below.
- **Press stroke:** the closed right gripper presses down on a laid strip and draws along it in one continuous stroke from end to end, without lifting off.
- **Press:** the closed right gripper set down flat on the applied label, held still for 2 seconds, and lifted straight off. A press never slides and never rubs.
- **Guided slide: Unvalidated at the station.** Both grippers closed before contact. The right gripper presses against the middle of the carton's right wall and supplies a slow push toward the lane spot. The left gripper rests against the middle of the left wall and travels with the carton as a guide, keeping it from turning. The base stays flat on the desk for the whole move, and neither gripper closes around the carton.
- **Facing:** the carton's near face pointing at the front edge. The carton keeps its facing from the packing spot to the lane spot.
- **Settled:** the item, ball, or carton stays put for 2 seconds after the gripper lifts away, with nothing springing back, rocking, or sliding.
- **Release point:** the inside of the carton, the crumple spot, the carton's top face, or the scrap cup. A gripper never opens anywhere else — in particular, never over the staging board or over the bare tabletop.

Steps

Steps 1 through 7 pack one order. Items go in in weight order, the far gap is filled before the near gap, and everything is lowered until it touches down before the gripper opens. Only the item picks in Steps 1 and 2 depend on the config: the **right gripper** takes each item from the back-center behind the carton (Config M), from the staging board (Config R1), or from the back-right (Config R2). Every other line is the same in all three configs.

Step 1: Load the base item

Goal: item 1, the heaviest, lies flat and seated on the carton floor with a clear gap at the far wall and at the near wall.

- The **left gripper**, closed, pins the **left wall of the carton** and holds it there for the whole step.
- **IF the items are at the back-center (Config M):** the **right gripper** takes **item 1**, the one nearest the carton's far wall, by its two long sides, lifts it flat off its spot, and carries it flat forward over the far wall to the carton.
- **IF the items are on the staging board (Config R1):** the **right gripper** takes **item 1**, the one nearest the packing spot, by its two long sides, lifts it flat off its spot, and carries it flat left to the carton.
- **IF the items are at the back-right (Config R2):** the **right gripper** takes **item 1**, the one nearest the packing spot, by its two long sides, lifts it flat off its spot, and carries it flat forward and left to the carton.

Then, in all three:

- Lower it onto the middle of the carton floor until it touches down, then open the gripper and lift clear.
- The **right gripper**, closed, presses the item down onto the floor at its **far edge**, then at its **near edge**, so it lies flat.

Carry the item flat the whole way. An item tipped on edge in the air has to be turned back down inside the carton, which scrapes it along a wall.

Check: item 1 lies flat and seated on the carton floor, not on edge and not leaning on a wall, with a clear gap along the far wall and a clear gap along the near wall. If it lands on edge, tilted, or across a gap, lift it straight out, lower it in again, and let it settle.

Step 2: Nest the three lighter items

Goal: items 2, 3, and 4 lie flat on the stack in weight order, with the top of the load below the carton rim.

- The **left gripper**, closed, keeps its pin on the **left wall of the carton**.
- **IF Config R1 or R2:** the **right gripper** takes each item off the row on the right, nearest the packing spot first. **IF Config M:** the **right gripper** takes each item off the row behind the carton, nearest the far wall first, and carries it forward over the far wall.
- The **right gripper** takes **item 2** by its two long sides, lifts it flat, and lowers it onto item 1 until it touches down. Open the gripper and lift clear.
- The **right gripper** takes **item 3** the same way and lowers it flat onto item 2 until it touches down. Open the gripper and lift clear.

- The **right gripper** takes **item 4** the same way and lowers it flat onto item 3 until it touches down. Open the gripper and lift clear.

Take the items in their staged order, closest to the packing spot first. Reaching past a nearer item for one further out knocks the row over.

Check: all four items lie flat on their largest faces in a stack, heaviest on the floor and lightest on top, none standing on edge, none leaning on a carton wall, and the top of the load sits below the carton rim. If an item stands on edge or leans, lift it straight out and lower it in again. If the load rides above the rim, lift the stack out, reseal item 1 flat, and load it again.

Step 3: Fill the voids with kraft paper

Goal: a paper ball squashed into the far gap, a paper ball squashed into the near gap, and a third ball pressed flat over the top of the load, all below the carton rim.

3.1 Make a paper ball

- The **right gripper** pinches the top **kraft sheet** by its near edge, lifts it off the paper stack, carries it to the **crumple spot**, and lays it flat there. Release.
- The **left gripper**, closed, presses down on the sheet's **near edge** and holds it there as a brace.
- The **right gripper**, closed, presses down on the sheet's **far edge** and drags it toward the front edge, up against the left gripper, so the sheet bunches up between them.
- The **right gripper** lifts off, comes down on the **left edge** of the bunched sheet, and drags it right, up against the left gripper, bunching it again.
- The **right gripper** pinches the bunched paper at its middle and lifts it clear as one ball. The **left gripper** lifts clear.

Both bunching drags come in against the same brace. One drag alone leaves a folded sheet, not a ball, and a folded sheet slides straight down a gap instead of filling it.

Check: the paper comes off the crumple spot as one loose ball, roughly as wide as the gap it is going into, with no part of the sheet trailing flat under it. If it lifts as a folded sheet, lay it back on the crumple spot and bunch it again.

3.2 Fill the far gap

- Make a paper ball as in 3.1.
- The **left gripper**, closed, pins the **left wall of the carton** and holds it there.
- The **right gripper** lowers the ball into the **far gap** until it touches the carton floor, then opens and lifts clear.
- The **right gripper**, closed, presses straight down on the ball until its top is level with the top of the load.

3.3 Fill the near gap

- The **left gripper** lets go of the wall. Make a second paper ball as in 3.1.
- The **left gripper**, closed, pins the **left wall of the carton** again.
- The **right gripper** lowers the ball into the **near gap** until it touches the carton floor, then opens and lifts clear.
- The **right gripper**, closed, presses straight down on the ball until its top is level with the top of the load.

3.4 Cover the load

- The **left gripper** lets go of the wall. Make a third paper ball as in 3.1.
- The **left gripper**, closed, pins the **left wall of the carton** again.
- The **right gripper** lowers the ball onto the middle of the **top of the load** until it touches down, then opens and lifts clear.
- The **right gripper**, closed, presses the ball down flat across the top of the load, first at the far side, then at the near side, until the whole top of the fill sits below the carton rim.

Check: both gaps are filled from the carton floor up to the top of the load with no gap left open down a wall, the third ball lies pressed flat across the whole top of the load, and everything sits below the carton rim. If a gap is still open, make another ball and press it in. If the fill stands above the rim, press it down again — do not fold the flaps over it.

Step 4: Shake-test the pack

Goal: two rocks with nothing in the load moving, so the pack is known to be tight before it is taped.

- The **left gripper**, closed, stands against the bottom of the carton's **left wall** so the carton cannot slide.
- The **right gripper**, closed, presses on the **right wall** near its top and pushes left until the carton tips up onto its left bottom edge, about a gripper's width of daylight under the right side. Hold for 1 second, then ease it back level.
- Do a second rock the same way.
- Both grippers lift clear, and the carton is looked at from above with the top still open.

Rock the carton on the desk. Lifting it off the desk to shake it drops the load onto the flaps and tells you nothing about how the pack travels.

Check: after both rocks, every item is still flat and in the same place in the stack, both gap balls are still packed from the floor to the top of the load, and the top fill still lies flat across the load. If any item has slid, tipped, or opened a gap, make another paper ball as in 3.1, press it into the gap that opened, and run the shake test again. Nothing goes into the carton after the shake test passes.

Step 5: Fold and tape the top

Goal: the four top flaps are folded shut and one tape strip runs down the center seam, pressed flat with both overhangs down.

5.1 Fold the flaps shut

- The **left gripper**, closed, pins the **left wall of the carton** and holds it there for the whole step.
- The **right gripper** folds the **right end flap** in, holds it flat for 2 seconds, and releases.
- The **right gripper** folds the **left end flap** in the same way.
- The **right gripper** folds the **far long flap** in over the end flaps, holds it flat for 2 seconds, and releases.
- The **right gripper** folds the **near long flap** in the same way, so the two long flaps meet at the center seam.

End flaps first, then long flaps. A long flap folded first ends up on top of the end flap and the seam will not close.

5.2 Lay and press the strip

- The **right gripper** pinches the **grab tab** of the tape strip, lifts it straight up off the peel board until it hangs straight and clear, and carries it to the carton without turning it.
- Lower the strip's free end onto the **center seam** at the left, so the free end sticks out past the left end panel. Travel right along the seam, letting the strip settle onto it behind the gripper, until the tab end sits past the right end panel. Release the tab.
- The **right gripper**, closed, presses down on the strip at the left overhang and draws along the seam in one continuous stroke, left to right, all the way to the right overhang, without lifting off.
- The **right gripper** presses the **left overhang** down onto the left end panel, then the **right overhang** down onto the right end panel.
- Both grippers lift clear.

Check: all four flaps lie flat with no corner standing up and nothing from the load caught in the seam, the strip runs down the middle of the center seam from end to end covering it along its whole length, it lies flat with no bubbles, folds, or wrinkles, and both overhangs lie flat against their end panels. If the strip landed off the seam, folded on itself, or lifted at an edge, peel it off, drop it in the scrap cup, and lay the spare strip. Never peel a strip up and stick it back down. Once the top is taped, the carton is not opened again.

Step 6: Apply the shipping label

Goal: the shipping label lying square inside the label patch on the top face, pressed flat.

- The **left gripper**, closed, pins the **left wall of the carton** and holds it there for the whole step.
- The **right gripper** pinches the label's **grab tab**, lifts it straight up off the peel board until it hangs straight and clear, and carries it to the top face without turning it.
- Lower the label straight down into the **label patch**, with its edges in line with the patch's edges, until it is lying on the flap. Release the grab tab.
- The **right gripper**, closed, presses down flat on the label, holds for 2 seconds, and lifts straight off.
- Both grippers lift clear.

Line the label up over the patch before you lower it. Adhesive sticks once, so a label is placed by lowering it, never by sliding it into line after it has touched the carton.

Check: the label lies inside the label patch, square, with no corner overhanging the outline, no corner lifted, no fold, and no bubble, and it is clear of the tape strip. If it lands off the patch, crooked, folded on itself, or on the tape, peel it off, drop it in the scrap cup, and apply the reserve label. Never peel a label up and stick it back down.

Step 7: Stage the carton to the lane

Goal: the sealed carton resting on the lane spot, facing the front edge, with its label up and readable.

- Close both grippers before either one touches the carton.
- The **right gripper** presses against the middle of the carton's **right wall** and supplies the push. The **left gripper** rests against the middle of its **left wall**, clear of the top face and the label.
- Make one slow **guided slide** straight left until the carton is fully inside the **lane spot**. The right gripper supplies the push; the left gripper travels with the left wall and keeps the carton square.
- Stop once the carton is centered in the lane spot, then lift both grippers straight clear.

This is the first and only time the carton is moved. Its base stays on the desk, its facing does not change, and neither gripper pinches, lifts, or closes across the carton.

Check: the carton rests flat and square inside the lane spot, settled, its near face toward the front edge, its label up, square, and readable from above, and the packing spot is bare. If the carton stops short of the lane spot or starts to turn, slide it back to the packing spot and make the guided slide again.

Step 8: End the episode

- Confirm the carton sits square on the lane spot with its top taped and its label flat on the patch, the packing spot bare, the start zone and the staging board bare, the crumple spot

bare, and the backing strip in the scrap cup.

- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the packed carton off the lane spot and move it to the outbound stack, off the tabletop. A packed carton is never unpacked and reused in the same session.
2. Empty the scrap cup and clear the tabletop of backing strips, scrapped labels, and loose paper.
3. Put a fresh taped carton square on the packing spot, open end up, with its four top flaps folded out flat.
4. Weigh the next order's four items and lay them out in a row in the start zone for the next episode's config — back-center behind the carton (Config M), on the staging board (Config R1), or back-right behind the staging board (Config R2) — in weight order, closest to the packing spot heaviest, each flat on its largest face and numbered 1 to 4.
5. Check that item 1 covers the carton floor and that no item is larger than the one before it.
6. Reload the rest of the staging board in the order it is used: a stack of four kraft sheets on the paper spot, then the peel board with one tape strip and one spare, and one shipping label and one reserve, all grab tabs pointing right.
7. Check the label patch on the new carton's flap is clean, dry, and bare.
8. Wipe the crumple spot, the packing spot, the lane spot, the start zone, and the staging board clear and dry.
9. Replace anything damaged — a crushed, split, or damp carton, a carton whose bottom tape has lifted, a torn kraft sheet, an item with a split seal, a strip or label that has lost its tack, or a label stuck to the peel board so it will not lift in one piece.
10. Check that the staging board, the peel board, and the scrap cup are firm and have not shifted.
11. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the start position: the violations below were written for Config R1 (the four items in a row on the staging board on the right). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the items are not in the start zone for the config; a gripper reaches across the table for an item; the left gripper takes an item because the row is nearer to it; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the right gripper takes every item from the start zone in all three configs; no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the items are before the first reach, then follow that config's IF lines through Steps 1 and 2.

Violation: Wrong order

- **Visible cue:** an item goes in after the fill has started, a gap is filled before all four items are in, the shake test is run before the top fill is pressed down, a flap is folded before the shake test, the label goes on before the tape strip is down, or the carton is slid to the lane before it is labelled.
- **SOP rule broken:** Steps 1 to 7 — nest, fill, shake, tape, label, stage, and each stage starts only after the one before it is complete.
- **Coaching note:** nest, fill, shake, tape, label, stage. Nothing jumps the queue.

Violation: Items loaded out of weight order

- **Visible cue:** the items are taken off the staging board out of their numbered order, an item is lowered onto a lighter one already in the carton, or the arm reaches past a nearer item for one further out.
- **SOP rule broken:** Steps 1 and 2 — load items 1 to 4 in their staged order, heaviest on the carton floor and lightest on top.
- **Coaching note:** nearest first, every time. The staged order is the weight order — do not re-sort it at the carton.

Violation: Item not laid flat or not seated

- **Visible cue:** an item lands on edge, tilted, leaning on a carton wall or on a neighbour, bridging a gap, or with a corner riding up a wall, and the episode moves on.
- **SOP rule broken:** Steps 1 and 2 — each item lies flat on its largest face on what is under it, and item 1 lies seated on the carton floor.
- **Coaching note:** flat on its big face, every time. A leaning item lifts the whole stack.

Violation: Item or ball dropped in instead of lowered

- **Visible cue:** the gripper opens above the carton and an item or a paper ball falls the last part of the way, bounces, or lands askew.
- **SOP rule broken:** Steps 1, 2, and 3 — lower each item and each ball until it touches down, then open the gripper.
- **Coaching note:** touch down first, then let go.

Violation: Arms swapped roles

- **Visible cue:** the left gripper lifts an item, a paper ball, the tape strip, or the label, folds a flap, or makes a press stroke; the right gripper pins or braces while the left one works; or either arm crosses to the other side of the tabletop.
- **SOP rule broken:** Steps 1 to 7 — the left arm holds and the right arm works, and the left gripper never carries an item, a paper ball, the tape strip, or the label.
- **Coaching note:** left holds, right works. The left gripper stays on the carton wall and on the sheet at the crumple spot.

Violation: Worked without a hold

- **Visible cue:** an item or ball is lowered in, a ball is pressed down, a flap is folded, a strip is laid, or a label is pressed while the left gripper is off the carton; a sheet is bunched with no left gripper brace on it; or the left gripper takes its pin only after the motion has started, or comes off before the step is finished.
- **SOP rule broken:** Steps 1 to 6 — the pin or brace is taken first and kept for the whole motion.
- **Coaching note:** hold first, let go last.

Violation: Near gap filled before the far gap

- **Visible cue:** a paper ball goes into the near gap while the far gap is still empty.
- **SOP rule broken:** Step 3 — fill the far gap first, then the near gap, then the top.
- **Coaching note:** far first, every time. Fill the near gap last so you never reach over it.

Violation: Paper put in without being made into a ball

- **Visible cue:** a flat or folded sheet is lowered into a gap or laid over the load, a sheet is pushed into a gap straight from the paper stack, or paper is bunched somewhere other than

the crumple spot.

- **SOP rule broken:** Step 3.1 — every sheet is bunched into a loose ball at the crumple spot with two drags against the left gripper brace before it goes into the carton.
- **Coaching note:** a folded sheet slides to the bottom of the gap and fills nothing. Bunch it twice, then lift it as one ball.

Violation: A void left unfilled

- **Visible cue:** a gap is left open down a wall, a ball sits at the bottom of a gap with the top of the gap still empty, a ball is put in but not pressed down, or the load is taped with no top fill over it.
- **SOP rule broken:** Step 3 — both gaps are filled from the carton floor up to the top of the load, and the third ball is pressed flat across the whole top.
- **Coaching note:** press each ball down until it fills the gap. Two gaps and the top, every time.

Violation: Fill or load left proud of the rim

- **Visible cue:** the top of the load or the top fill rides above the carton rim and the flaps are folded over it anyway, or a flap is pushed down onto the fill to make it close.
- **SOP rule broken:** Steps 2, 3, and 5 — the top of the load sits below the rim before the flaps are folded.
- **Coaching note:** if it stands above the rim, press the fill down first. Do not force the flaps over it.

Violation: Shake test skipped or cut short

- **Visible cue:** the flaps are folded with no rock at all, only one rock is done, a rock is not held at the tilt, the carton is lifted off the desk to shake it, or the carton is not looked at from above after the rocks.
- **SOP rule broken:** Step 4 — two rocks on the desk, each held for 1 second, then a look down into the open carton.
- **Coaching note:** two rocks, on the desk, then look inside. The test is the look, not the wobble.

Violation: Shake test failed and not corrected

- **Visible cue:** an item slides, tips, or a gap opens during the rocks, and the episode goes straight on to folding the flaps without another ball being pressed in and the rocks repeated.
- **SOP rule broken:** Step 4 — if anything moves, press another ball into the gap that opened and run the shake test again.
- **Coaching note:** movement means the pack is loose. Fill it and rock it again before you tape.

Violation: Placement corrected by shoving

- **Visible cue:** a gripper nudges, pats, or slides an item, a paper ball, a flap, the strip, or the label into position after releasing it, instead of making the correction the SOP calls for.
- **SOP rule broken:** Steps 1 to 7 — correct an item or a ball by lifting it straight out and lowering it in again, and correct a strip or a label by scrapping it and taking a fresh one.
- **Coaching note:** do it again properly, or leave it alone.

Violation: Flaps folded in the wrong order or not flat

- **Visible cue:** a long flap is folded in before both end flaps, a flap is left standing up or folded back on itself, a fold is released before it has been held flat, or paper or an item is caught in the center seam.
- **SOP rule broken:** Step 5.1 — right end flap, left end flap, far long flap, near long flap, each held flat for 2 seconds.
- **Coaching note:** end flaps first, long flaps over them. Hold each fold before you let go.

Violation: Strip taken by the wrong part or backing mishandled

- **Visible cue:** the right gripper pinches the strip anywhere other than its grab tab, touches the sticky side, carries a strip that is curled or stuck to itself instead of scrapping it, or drops the backing strip on the tabletop or into the carton instead of the scrap cup.
- **SOP rule broken:** Step 5.2 — the grab tab is the only part of a strip a gripper touches, a strip that does not hang straight is scrapped, and backing goes in the scrap cup.
- **Coaching note:** tab only, and let it hang before you carry it. A strip that has touched the gripper is already ruined.

Violation: Tape laid off the center seam

- **Visible cue:** the strip lands off the seam, crooked to it, short of an end panel at either end, or folded on itself, and the episode moves on.
- **SOP rule broken:** Step 5.2 — lower the free end onto the center seam at the left and travel right so the strip settles down the middle of the seam with an overhang past each end panel.
- **Coaching note:** put the free end on the seam before you start moving. The first end decides where the whole strip goes.

Violation: Tape press incomplete

- **Visible cue:** the episode moves on from a laid strip with no press stroke, the gripper lifts partway along and sets down again, the strip is dabbed at in several places, or an overhang is left standing off its end panel.
- **SOP rule broken:** Step 5.2 — one continuous press stroke along the seam end to end, then both overhangs pressed down onto their end panels.
- **Coaching note:** one stroke, all the way across, then finish both overhangs.

Violation: Tape or label peeled up and stuck down again

- **Visible cue:** a strip or a label that is already down gets lifted off and pressed back on, shifted after touching down, or repositioned by peeling one end.
- **SOP rule broken:** Steps 5.2 and 6 — a badly laid strip or label is peeled off, dropped in the scrap cup, and replaced with the spare.
- **Coaching note:** adhesive sticks once. Throw it away and take a fresh one.

Violation: Label laid off the patch

- **Visible cue:** the label lands outside the label patch, crooked to it, overhanging its edge, on the tape strip, on another face of the carton, folded on itself, or with a bubble or a lifted corner left unpressed.
- **SOP rule broken:** Step 6 — lower the label straight down into the patch with its edges in line with the patch's edges.
- **Coaching note:** line it up over the patch before you lower it. Sliding it afterwards is not an option.

Violation: Label press skipped

- **Visible cue:** the episode moves on from an applied label with no press, the press is held for less than 2 seconds, or the closed gripper slides or rubs across the label instead of pressing straight down.
- **SOP rule broken:** Step 6 — press down flat on the label, hold for 2 seconds, and lift straight off.
- **Coaching note:** straight down, hold, straight up. A press never rubs.

Violation: Carton moved off the packing spot early

- **Visible cue:** the carton is lifted, turned, dragged, or pushed off the packing spot at any point before Step 7 — including being pulled along by a press, a fold, or a rock.
- **SOP rule broken:** Steps 1 to 6 — the carton stands square on the packing spot until it is sealed and labelled.
- **Coaching note:** pin the wall so your own push does not move the carton.

Violation: Carton staged to the lane wrong

- **Visible cue:** a gripper closes around the carton, pinches it, or lifts it to move it; the carton is pushed with an open gripper; it turns during the move so its facing changes; it stops short of the lane spot or overshoots it; or it is left sitting across the outline.
- **SOP rule broken:** Step 7 — one slow guided slide with both grippers closed, the right gripper pushing and the left gripper guiding, until the carton is centered in the lane spot with its facing unchanged.
- **Coaching note:** closed grippers, base on the desk, one slow slide. Push with the right, guide with the left.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the staging board, over the bare tabletop, or over open space, and an item, a paper ball, a strip, a label, or the backing lands anywhere other than in the carton, on the crumple spot, on the carton's top face, or in the scrap cup.
- **SOP rule broken:** Steps 1 to 7 — a gripper opens over the carton, the crumple spot, the top face, or the scrap cup, and nowhere else.
- **Coaching note:** know where it will land before you open the gripper.

Violation: Item dropped, collided, or knocked something over

- **Visible cue:** an item, a paper ball, a strip, or a label falls on the tabletop or the floor; the staged row of items or the paper stack is knocked over; the peel board or the scrap cup is struck and shifts; or the two arms hit each other over the carton.
- **SOP rule broken:** Steps 1 to 7 — keep the travel path clear, and only release once the piece has settled.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the top untaped, an overhang unpressed, the label missing or unpressed, the carton still on the packing spot or across the lane outline, an item or sheet still on the staging board, paper left on the crumple spot, the backing strip on the tabletop, or an arm away from home.
- **SOP rule broken:** Step 8 — confirm the taped and labelled carton on the lane spot, the bare packing spot, staging board, and crumple spot, and the backing in the scrap cup, then return both arms home, then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, a controller-caused collision, or a motor error.
- **Defective carton:** crushed, split, damp, out of square, or a bottom seam whose tape has lifted so the floor sags. Replace it before the next episode.
- **Defective item:** a split seal, a leaking or damaged item, or an item that will not stand flat on its largest face.

- **Items sized wrong for the job:** item 1 does not cover the carton floor, an item is larger than the one before it, or the four stacked leave no room for the top fill below the rim.
- **Defective kraft sheet:** torn, damp, or too small to make a ball that fills a gap.
- **Defective strip or label:** lost its tack, torn while being lifted, or stuck to the peel board so it will not lift in one piece.
- **Staging board, peel board, or scrap cup shifts out of position** during the episode.

Annotation subtasks (from SOP)

1. Lower one item flat into the carton
2. Bunch one kraft sheet into a paper ball at the crumple spot
3. Press one paper ball down into a gap
4. Press the top fill flat over the load
5. Rock the carton once for the shake test
6. Fold one top flap in
7. Lay the tape strip on the center seam and press it down
8. Apply the shipping label to the patch and press it
9. Slide the carton to the lane spot with both grippers
10. Return both arms home and end the episode